

Problem 1

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Problem 1 Fill in the following blanks with the correct choice of the words from this list:

Increasing, decreasing, positive, negative, concave up, concave down

- (a) If you know that $f''(x) > 0$, for all x in some open interval I , then you know $f'(x)$ is _____ and $f(x)$ is _____, for all x in I .

Explanation. increasing, concave up

- (b) If you know $g'(x) < 0$ and decreasing, for all x in some open interval I , then you know $g(x)$ is _____ and _____, for all x in I .

Explanation. decreasing, concave down

- (c) If you know $h(x)$ is positive, increasing, and concave down, for all x in some open interval I , then you know $h'(x)$ is _____ and _____ and that $h''(x)$ is _____, for all x in I .

Explanation. positive, decreasing, negative